



Rational Decision Making

Decision Theory, Utility Theory, and Game Theory

CS4246/5446

Reinforcement Learning and Sequential Decision Making

Version 3.0



Topics

- Decision Making under Uncertainty
 - Rational decision making
 - Foundations of Decision Theory
 - Maximum Expected Utility Principle
- Basis of Utility Theory
 - Axioms of Rational Preferences
 - Expected Utility Theorem
 - Utility Functions



Decision Making under Uncertainty

Uncertain states and action effects



Rational Decision Making

- **Context:**
 - *Decision making under uncertainty*, in episodic and partially observable environments.
- **Core Idea:**

A rational agent combines:

 - Beliefs – information about the world
 - Preferences – values or goals
 - A decision process – selects actions that maximize expected benefit or minimize cost, within resource limits
- **Types of Rationality:**
 - Substantive (Ends-focused): Achieving desirable outcomes
 - Procedural (Mean-focused): Using a coherent, justifiable method
 - Meta-level: Deciding how to decide (bounded rationality, meta-reasoning)



The BDI Model in Agentic AI

- **Belief–Desire–Intention (BDI) Framework:**

- Beliefs: The agent's model of the world
- Desires: Goals or preferred outcomes
- Intentions: Committed plans or actions

- **In Agentic AI Systems (e.g., LLM agents):**

BDI provides a natural model for rational behavior:

- Beliefs inform decision-making
- Desires guide prioritization
- Intentions anchor planning and execution

Modern use: Characterizes AI agents that combine reasoning, planning, and tool use.

Decision Making under Uncertainty

- Decision Model:

- **Actions**: $a \in A$
- **Uncertain current state**: $s \in S$ with probability of reaching: $P(s)$
- **Transition model** of uncertain action outcome or effects:
 $P(s'|s, a)$ – probability that action a in state s reaches state s'
- **Outcome** of applying action a :
Result(a) – random variable whose values are outcome states
- **Probability of outcome state** s' , conditioning on that action a is executed:
 $P(\text{Result}(a) = s') = \sum_s P(s)P(s'|s, a)$
- **Preferences** captured by a **utility function**:
 $U(s)$ – assigns a single number to express the desirability of a state s



From Logical Planning to Decision Modeling

- Key Points:

- Logical planning:
the basis → states as propositions, actions as rules, goals as logical formulas.
- Decision models:
add probability + utility → handle uncertainty and preferences.
- Classical planning is thus a *special case* of decision-theoretic planning: deterministic transitions + binary utilities.

Types of Decision Theory

- Normative decision theory
 - Defines how ideal agents *should* decide
- Descriptive decision theory
 - Describes how real agents *actually* decide
- Prescriptive decision theory
 - *Guides* rational decision making in real-world settings.



The Beginnings

- **Utility Theory**

- Origin: Daniel Bernoulli (1738) — measuring risk with subjective value (utility).
- Scope: Single-agent decision making under uncertainty.

- **Game Theory**

- Origin: Von Neumann & Morgenstern (1944) — strategic interaction among agents.
- Scope: Multi-agent decision making, where outcomes depend on all players' actions.

Maximum Expected Utility (MEU) Principle

- **Fundamental Assumption:**

Rational agents choose actions that maximize **expected utility**.

$$\text{action} = \arg \max_a \mathbb{E}[U(a)] = \arg \max_a \sum_{s'} P(\text{Result}(a) = s') U(s')$$

Or, incorporating uncertainty in current state:

$$\mathbb{E}[U(a)] = \sum_{s'} \underbrace{\sum_s P(s)}_1 P(s'|s, a) U(s')$$

- Where:
- $P(s)$: belief about current state **1**
- $P(s'|s, a)$: transition model
- $U(s')$: utility of resulting state

= 1
for fully observable environment



Example: Panda's Simple Lunch Choice

Utility Theory in Action

- **Scenario:**
 - Panda bear Lulu must choose where to find lunch today.
- **Beliefs:**
 - Bamboo Grove: 50% chance of finding 10 bamboo shoots, 50% chance of finding none.
 - Berry Bush: 100% chance of finding 4 berries.
- **Preferences:**
 - Assume equal preferences for bamboo shoots and berries.
 - More food = higher utility (utility equals number of food items).

Decision:
Where should Lulu go for lunch?



Example: Pandas' Dilemma

Game Theory in Action

- **Scenario:**

Two pandas, Bobo and Almo, find bamboo in short supply.

Each day, they must decide whether to:

- **Share** bamboo grove locations
- **Keep** them secret

	<i>A: Keep</i>	<i>A: Share</i>
<i>B: Keep</i>	$B = +1; A = +1$	$B = +5; A = 0$
<i>B: Share</i>	$B = 0; A = +5$	$B = +3; A = +3$

- **Payoffs:**

- Both Keep: each gets +1 (safe but poor outcome)
- One Shares, One Keeps: the keeper gains +5, the sharer gets 0
- Both Share: both get +3 (cooperative outcome)

- **Decision:**

- What should the pandas do to maximize happiness and health (by eating A LOT OF bamboo)?

What Should the Pandas Do?

	<i>A: Keep</i>	<i>A: Share</i>
<i>B: Keep</i>	$B = +1; A = +1$	$B = +5; A = 0$
<i>B: Share</i>	$B = 0; A = +5$	$B = +3; A = +3$

- **Rational Action (Self-Interest):**

- Both pandas will likely choose to Keep Bamboo Locations Secret, leading to an optimal policy (equilibrium) with payoffs $[+1, +1]$.

- **Optimal Collective Action:**

- If they could build trust and cooperate to Share bamboo locations, both would benefit more $\rightarrow$ payoffs $[+3, +3]$.

- **Lesson:**

- Game theory shows that what is individually rational may not be collectively optimal.



Notes: Common Applications

Domain	Utility Theory Applications (Single-agent focus)	Game Theory Applications (Multi-agent focus)
Economics	Consumer choice, cost–benefit analysis, risk assessment	Pricing strategies, market competition, auctions
Networks	Resource allocation, service optimization	Cooperation, cost-sharing, competitive network use
Political Science	Policy evaluation, welfare optimization	Voting systems, coalition formation
Biology	Habitat choice, foraging optimization	Evolutionary stable strategies, predator–prey dynamics
Business	Product design trade-offs, investment decisions	Pricing competition, negotiations
Social Impact	Fair resource distribution, welfare maximization	Fairness and trust in strategic settings
Health Care	Treatment choice, cost-effectiveness analysis	Hospital competition, vaccine allocation games
Education	Curriculum optimization, personalized learning	Competition between institutions, collaboration incentives

Game Theory Reference: Maschler, Michael; Solan, Eilon; Zamir, Shmuel. Game Theory (p. xxiii). 2020. Cambridge University Press. Kindle Edition.



Axioms of Rational Preferences

Constraints on rational preferences

Rational Preferences

- Preference notations:

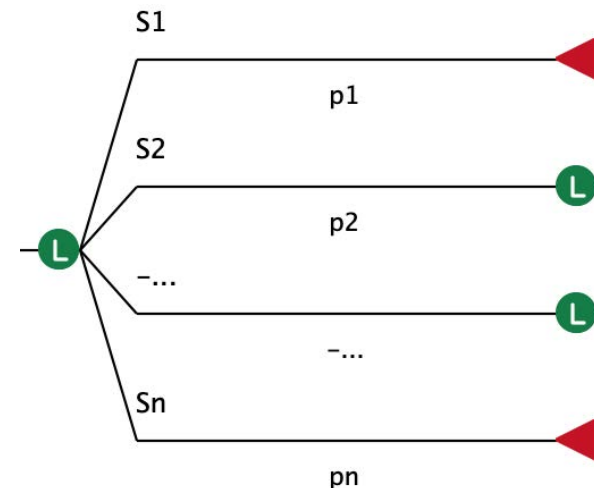
- $A \succ B$: Agent prefers A to B
- $A \sim B$: Agent is indifferent between A and B
- $A \succsim B$: Agent prefers A over B or is indifferent

- Lotteries:

- An action can be seen as a lottery L over outcomes
- Outcomes: $S_1, \dots, S_n$
- Probabilities: $p_1, \dots, p_n$
- Representation:

$$L = [p_1, S_1; p_2, S_2, \dots, p_n, S_n]$$

- Each S_i is a possible next state (from one of the action's possible effects)
- Outcomes may be atomic states or nested lotteries





Axioms of Rational Preferences

- Utility Theory:

- Explains how preferences over lotteries relate to preferences over the underlying states.

- Key Idea:

- Preferences must satisfy certain rationality conditions.
- If any axiom is violated, the agent can behave **irrationally** in some situations.

- Six axioms (or conditions of rationality):

- A1: Orderability (Completeness)
- A2: Transitivity
- A3: Continuity
- A4: Substitutability (Independence)
- A5: Monotonicity
- A6: Decomposability



A1. Orderability (Completeness)

- Definition:

- For any two lotteries A and B , a rational agent must:
- Prefer $A \succ B$, or
- Prefer $B \succ A$, or
- Be indifferent $A \sim B$.
- Intuition: *Any two options can be compared: you must prefer one or be indifferent.*

- Panda Story:

- Lulu the panda must decide: bamboo, berries, or no preference.

- Example:

- Choose between:
- $A = 5$ bamboo shoots
- $B = 1$ bowl of berries



A2. Transitivity

- Definition:

$$A \succ B \text{ and } B \succ C \implies A \succ C$$

- Intuition: *Preferences must be consistent: if A is better than B, and B is better than C, then A is better than C.*

- Panda Story:

- If Lulu prefers bamboo over berries, and berries over honey, she must also prefer bamboo over honey.

- Example:

- Bamboo $\succ$ Berries $\succ$ Honey $\rightarrow$ Bamboo $\succ$ Honey

A3. Continuity

- Definition:

$$A \succ B \succ C \implies \exists p [p, A; 1 - p, C] \sim B$$

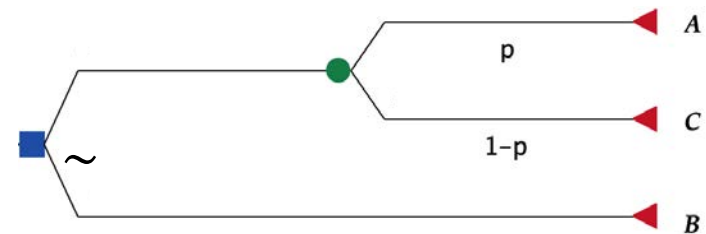
- Intuition: *A middle option is equivalent to some gamble between the best and the worst.*

- Panda Story:

- Lulu might be indifferent between “5 shoots for sure” and “a gamble: 10 shoots with 60%, 1 shoot with 40%.”

- Example (Lottery):

- Guaranteed 5 bamboo shoots (B)
- Or a lottery: 60% chance 10 shoots (A), 40% chance 1 shoot (C)



A4. Substitutability (Independence)

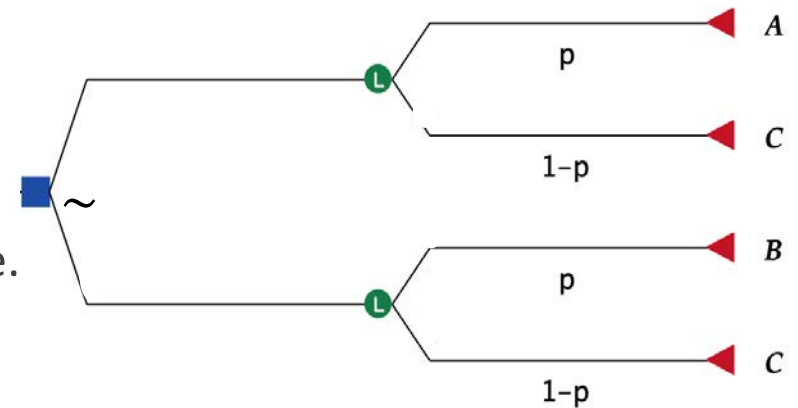
- **Definition:** OR $A \succ B \Rightarrow [p, A; 1 - p, C] \succ [p, B; 1 - p, C]$

$$A \sim B \Rightarrow [p, A; 1 - p, C] \sim [p, B; 1 - p, C]$$

- Intuition: *If two options are equally good, swapping one for the other in any lottery makes no difference.*

- **Panda Story:**

- If Lulu values bamboo and berries equally, swapping them in a lottery makes no difference.



- **Example (Lottery):**

- Indifferent between 5 shoots (A) and a bowl of berries (B)
- In a 70% lottery with nothing (C), Panda should be indifferent whether it's A or B.

A5. Monotonicity

- Definition:

$$A \succ B \implies (p > q \iff [p, A; 1 - p, B] \succ [q, A; 1 - q, B])$$

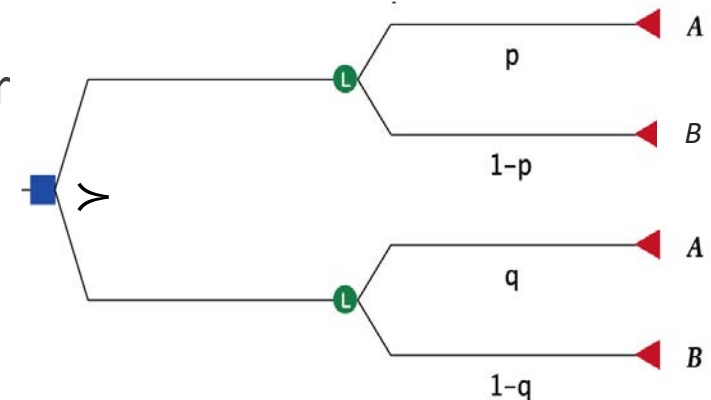
- Intuition: *A higher chance of a better outcome makes a lottery more desirable.*

- Panda Story:

- Since Lulu prefers bamboo, she should prefer the lottery with a higher bamboo chance.

- Example (Lottery):

- 80% chance bamboo, 20% berries
- 60% chance bamboo, 40% berries



A6. Decomposability

- Definition:

$$[p, A; 1 - p, [q, B; 1 - q, C]] \sim [p, A; (1 - p)q, B; (1 - p)(1 - q), C]$$

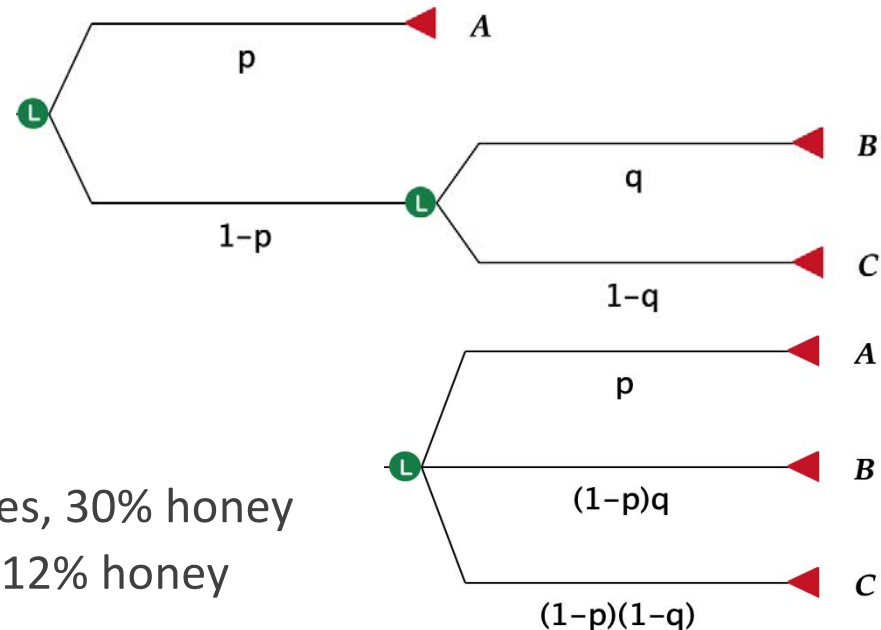
- Intuition: *Only final probabilities matter — multi-stage lotteries reduce to a single equivalent lottery.*

- Panda Story:

- From Lulu's view, it doesn't matter if the gamble has 2 stages or 1
— only the final probabilities count.

- Example (Lottery):

- 60% chance bamboo
- 40% chance → another lottery: 70% berries, 30% honey
- Equivalent to: 60% bamboo, 28% berries, 12% honey





Rational Preferences and Utility

- **Key Question:**
 - How should a rational agent make decisions?
- **Notes:**
 - Utility axioms = preference axioms
 - If preferences satisfy these axioms, a utility function U exists
- **Goal:**
 - Derive rational behavior from preference assumptions
- **Foundation:**
 - von Neumann–Morgenstern Expected Utility Theorem (1944)



Expected Utility Theorem

Von Neumann and Morgenstern (1944)

Existence of Utility Function

- If an agent's preferences obey the axioms of utility; then there exists a function U such that:
 - $U(A) > U(B)$ if and only if A is preferred to B , and
 - $U(A) = U(B)$ if and only if the agent is indifferent between A and B .

$$U(A) > U(B) \Leftrightarrow A \succ B \quad \text{and} \quad U(A) = U(B) \Leftrightarrow A \sim B$$

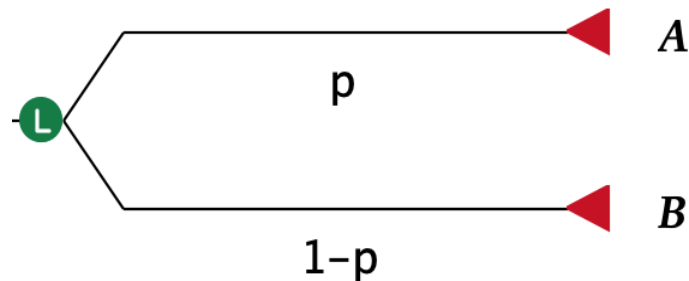
Expected Utility of A Lottery

- Utility of a Lottery:

- The utility of a lottery is the weighted average of the utilities of each possible outcome:

$$U([p_i, S_i; \dots; p_n, S_n]) = \sum_i p_i U(S_i)$$

- Once the probabilities and utilities of the outcomes are known, the utility of the entire lottery is fully determined.



$$U(L) = pU(A) + (1 - p)U(B)$$

Computing Expected Utility

- Expected Utility:

- When an action has uncertain outcomes, it behaves like a lottery.
- The expected utility of an action is the average utility of its possible outcomes, weighted by their probabilities.

$$EU(a) = \sum_{s'} P(\text{Result}(a) = s') U(s')$$

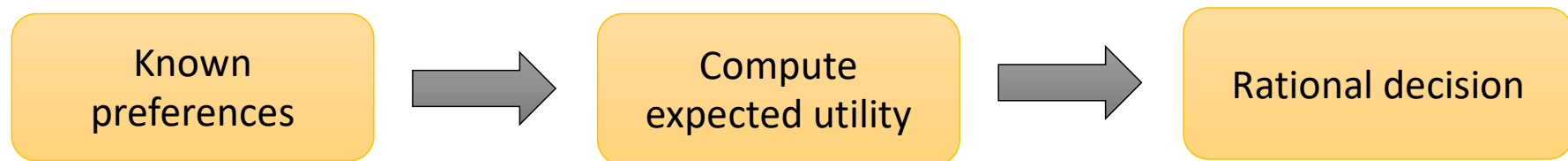
- Non-Unique Utility Functions:

- Utility is defined up to an **affine transformation**:

$$U'(s) = aU(s) + b \text{ with } a > 0$$

Stretch or shift the scale of utility, and the agent's decisions stay the same

Rational Decision Making: MEU Principle



- **Maximum expected utility (MEU) principle**

- Based on preferences, a rational agent chooses the action with the highest expected utility.

A prescription for intelligence behavior
– “do the right thing” – a basis for AI

- Choose action:

$$a^* = \operatorname{argmax}_a EU(a)$$

$$EU(a) = \sum_{s'} P(\text{Result}(a) = s') U(s') = \sum_{s'} \sum_s P(s) P(s'|s, a) U(s')$$



Computing MEU

- Key Challenges

- Must evaluate many possible actions
- Requires estimating probabilities: $P(s)$, $P(s'|s, a) \rightarrow$ needs perception, learning, inference, causal models
- Requires estimating utilities: $U(s') \rightarrow$ may involve planning or search
- Utility estimates may be uncertain

- Role in AI

- Decision theory provides the mathematical framework, but not a full AI solution.
- Utility functions act as internalized performance measures:
 - Performance measure $\rightarrow$ evaluates complete histories (external view)
 - Utility function $\rightarrow$ guides choices step by step (internal view)



Rational Decision-Theoretic Agent

- Idealization:
 - Assumes an **ideal**, hypothetical and fully rational agent
- Preferences:
 - Satisfy rationality axioms
- Rationality:
 - Chooses the most preferred (optimal) option
- MEU Principle:
 - Guides decision making and problem solving



Utility Functions

Encoding Preferences in Utility Functions

Utility Functions in Real-World Domains

Utility of Money and Risk Attitudes

Encoding Preferences in Utility Functions

- **Utility Function:**

- A **utility function** $U(x)$ maps outcome desirability into numeric utility values
- x : Cost, profit, life expectancy, jobs saved, etc.
- Captures trade-offs, value judgments, and risk attitudes

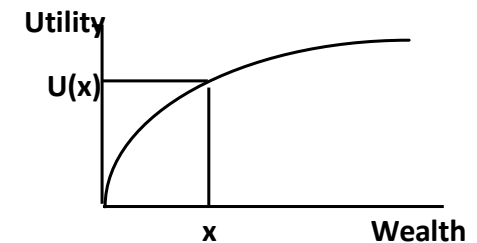
- **Why use utility functions?**

- Compare and rank outcomes
- Support decision-making under uncertainty
- Aggregate multiple aspects of value (e.g., cost, health, time)

- **Assessment methods:**

- Direct: certainty equivalents, probability equivalents
- Indirect: expert consensus, published data
- Use multi-attribute utility functions for complex trade-offs

$$U(x) = \log(x)$$
$$U(x) = 1 - e^{-x/R}$$
$$U(x) = x^{0.5}$$



Wealth (x)	Utility Value U(x)
2500	1.50
1500	1.24
1000	0.93
600	0.65
400	0.47
0	0.15

Notes: Utility Functions in Real-World Domains

Domain	Utility Function / Measure	Description
Health	QALY (Quality-Adjusted Life Year)	$U = \text{Years} \times \text{Quality score (0-1)}$ e.g., 10 years at 0.6 quality = 6 QALYs
	DALY (Disability-Adjusted Life Year)	$U = \text{Years lost to disability or death}$ Lower DALY = better outcome
	Micromort	$U = -\text{Risk in 1M chance of death}$ 1 micromort = 10^{-6} probability of death
Public Policy	VSL (Value of Statistical Life)	$U = -\text{Monetary cost per statistical life saved}$ e.g., \$10M/life used in EPA, FDA and DOT
Transportation	VHT (Value of Hour Traveled)	$U = -\text{Time cost} \times \text{Value per hour}$ e.g., \$20/hr for commuters
Finance	Exponential Utility	$U(x) = 1 - e^{-rx}$ Captures risk aversion ($r > 0$)
Economics	Log Utility	$U(x) = \log(x)$ Reflects diminishing marginal utility of income
Energy/Environment	Carbon cost utility	$U = -\text{tons CO}_2 \times \$ \text{ per ton (e.g., \$50)}$ Used in carbon pricing & offset decisions



Example: Value of Statistical Life

- Value of statistical life:
 - How to put a value on human life?
 - Used by agencies of U.S. government, including the Environmental Protection Agency, Health and Human Services, and the Department of Transportation
 - To determine costs and benefits of regulations and interventions
 - How much would it be?
 - USA: ~USD\$14 million (in 2025)
 - Australia: AUD\$5.9 million (in 2025)

Sources: (accessed 26 Aug 2024)

- [U.S. Department of Health and Human Services](#)
- [Australian Office of Impact Analysis](#)

Example: A Choice of Games

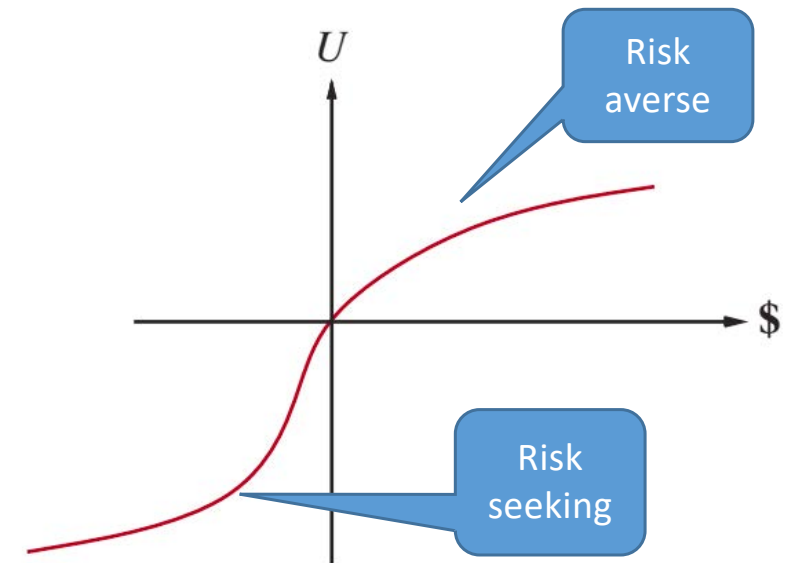
- Consider the following two games: EMV
 - Game 1 Win \$30 with prob 0.5
Lose \$1 with prob 0.5
 - Game 2 Win \$2000 with prob 0.5
Lose \$1900 with prob 0.5
- Which game would you play? Why?
- What if you are to play a game 10 times?

Example: Expected Monetary Value

- Example:
 - You have won \$1 mil so far
 - Now, flip a coin –
 - Lose all the money (new total \$0) if Head and
 - Gain \$1.5mil (new total \$2.5mil) if Tail
 - Will you play?
- What is the expected monetary value (EMV) of the lottery?
 - $EMV =$
- Is accepting the game the best decision?

Utility of Money

- Let S_n = state of having $\$n$; current wealth = $\$k$
- Expected Utility of lottery:
 - $EU(\text{Accept}) = \frac{1}{2}U(S_k) + \frac{1}{2}U(S_{k+2.5M})$
 - $EU(\text{Decline}) = U(S_{k+1M})$
- **Example utilities:**
 - $U(S_k) = 5$, $U(S_{k+2.5M}) = 9$, $U(S_{k+1M}) = 8$
 - $EU(\text{Accept}) = 0.5(5) + 0.5(9) = 7$
 - $EU(\text{Decline}) = 8$
 - Decision: Decline is better ($8 > 7$)
- **Note:**
 - The utility of money is usually concave (e.g., $\sim \log$ of wealth), reflecting risk aversion.



A typical utility of money function

Source: RN16.2 b

Expected Utility and Certainty Equivalent

- **Certainty Equivalent:**

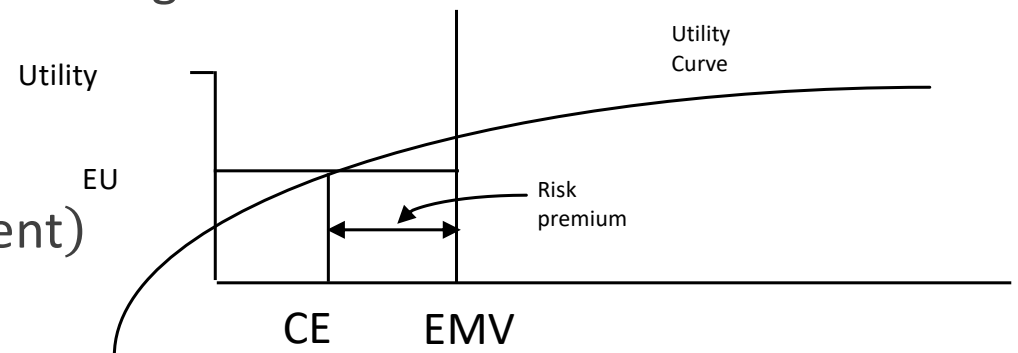
- The certainty equivalent of a lottery is the guaranteed outcome the agent considers equally desirable.

- **Defined by:**

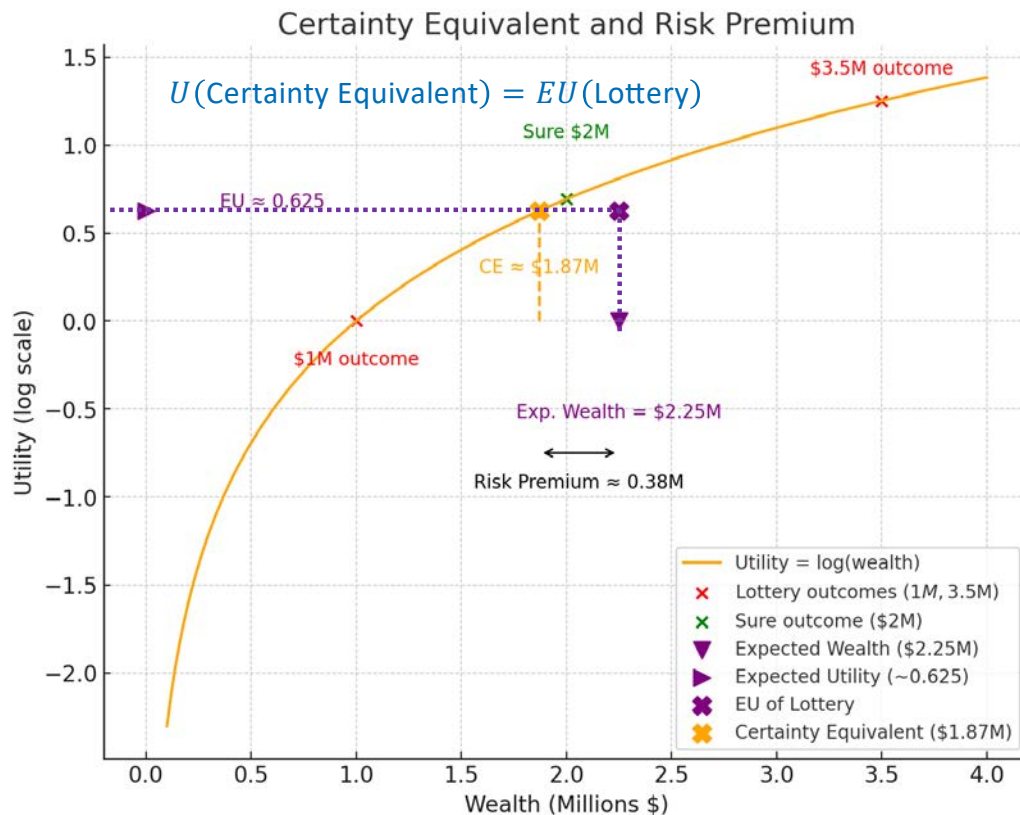
- $EU(\text{Lottery}) = U(\text{Certainty Equivalent})$

- **Key Points:**

- If two options have the same CE $\rightarrow$ they have the same expected utility $\rightarrow$ agent is indifferent.
- Ranking options by their CE is equivalent to ranking by expected utility.



Certainty Equivalent and Risk Premium



- **Orange curve:** log utility, concave, showing diminishing returns.
- **Red points:** risky lottery outcomes (\$1M and \$3.5M).
- **Green point:** sure outcome (\$2M).
- **Purple markers:**
 - **▼** Expected wealth (EMV = \$2.25M) - arithmetic average of the monetary outcomes (x-axis).
 - **►** Expected utility (~ 0.625) - arithmetic average of utilities of the outcomes (y-axis).
 - **X** EU of lottery- expected utility aligned to EMV for visualization.
- **Orange x :** certainty equivalent (CE $\approx$ \$1.87M)
 - $U(\text{Certainty Equivalent}) = EU(\text{Lottery})$
- **Black arrow:** Risk Premium ($\sim$ \$0.38M)

Risk Premium = EMV – CE : gap between expected wealth and CE, — the amount one would give up to avoid the risk.

Risk Attitudes and Risk Premium

- Risk Premium:

- Risk Premium = Expected Value (EV) – Certainty Equivalent (CE)
- Value given up (or paid) to avoid (or take on) risk.

- Risk Attitudes:

- Risk-averse: $CE < EV \rightarrow$ prefers a sure but lower payoff
- Risk-seeking: $CE > EV \rightarrow$ willing to pay for risky gain
- Risk-neutral: $CE = EV \rightarrow$ indifferent





Notes: Calculating Risk Premium

- **Definition:**

- Risk Premium = Expected Value (EV) – Certainty Equivalent (CE)
- $EU(\text{Lottery}) = U(\text{CE})$

- **Steps to Calculate:**

- Compute Expected Utility (EU) of the lottery.
- Find CE: the value whose utility equals the EU.
- Compute EV of the lottery (e.g., in money, time, health units).
- Calculate: Risk Premium = EV – CE

- **Notes:**

- EV = Expected Value in the original units (e.g., dollars, years, QALYs).
- For monetary outcomes, EV is called EMV.
- Utility scale must reflect preferences between best and worst outcomes.
- CE and PE (Probability Equivalent) methods can be used for utility assessment.

Exercise: Paying to Avoid Risk

- Recall the following gamble:
 - Win \$2000 with probability 0.5
 - Lose \$20 with probability 0.5
- What is the expected **value** of the lottery (EMV)?
 - $EMV = \$X$
- How much will you be willing to accept to avoid the lottery?
 - Let $CE = \$300$ (your subjective, “perceived value” of the lottery)
- What is the risk premium?
 - Risk premium = $\$X - \$300 = \$Y$
- What does this mean?
 - Trading lottery for \$300: willing to give up $\$X$ in expected value to **avoid risk**



Summary

- Preference & Utility

- Preferences: $A \succ B$ (preferred), $A \sim B$ (indifferent)
- Utility function $U(s)$: assigns numbers to outcomes, capturing ranking and risk attitude
- Subjective & personal: not directly comparable across individuals
- Ordinal scale: numbers express order, not strength of preference

- MEU Principle

- Rational choice = pick action with Maximum Expected Utility (MEU)
- Certainty Equivalent (CE): the sure value equal in utility to a lottery
- EU = utility scale; CE = real-world scale

- Caveats

- Utility is non-linear; not simply additive $U(a + b) \neq U(a) + U(b)$
- Utilities must be assessed at outcome endpoints
- Multi-attribute and uncertain cases need more advanced models



Rational Decision Making in the Real World

- **AI Applications**

- Decision-making under uncertainty with MEU
- Applied in planning, learning, and inference
- Supports human–AI collaboration and transparency

- **Challenges**

- Capturing human preferences in dynamic, uncertain contexts
- Integrating expert judgment with data
- Building responsible AI: fairness, ethics, and governance



Homework

- Readings:

- RN: 16.1: Maximum expected utility principle
- RN: 16.2-16.3.2: Utility Theory

- Reviews:

- Review notes on Probability and Statistics
- RN: 13.2-13.5; 14.2-14.4 (Conditional probability and Bayesian networks)
- Charniak, E., Bayesian networks without tears. *AI Mag.*, 1991. 12(4): p. 50–63.
- RN: 16.5 (Decision networks), 16.6.6 (Sensitivity analysis)
- Howard, R.A., [Decision Analysis: Practice and Promise](#). *Management Science*, 1988. 34(6): p. 679-695. [Accessible through NUS Library e-Resources]

- References

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